

Questions with Answer Keys

MathonGo

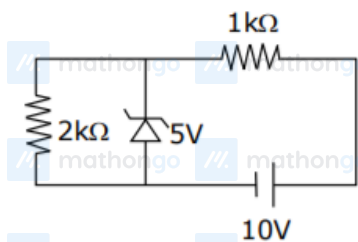
Q1: 24 Feb (Shift 1) - Single Correct

If an emitter current is changed by 4 mA, the collector current changes by 3.5 mA. The value of β will be -

- (1) 7
- (2) 0.875
- (3) 0.5
- (4) 3.5

Q2: 24 Feb (Shift 1) - Numerical

In connection with the circuit drawn below, the value of current flowing through $2k\Omega$ resistor is $____ \times 10^{-4}$ A



Q3: 24 Feb (Shift 2) - Single Correct

Zener breakdown occurs in a p-n junction having p and n both :

- (1) lightly doped and have wide depletion layer.
- (2) heavily doped and have narrow depletion layer.
- (3) heavily doped and have wide depletion layer.
- (4) lightly doped and have narrow depletion layer.

Q4: 24 Feb (Shift 2) - Single Correct

Given below are two statements :

Statement I : PN junction diodes can be used to function as transistor, simply by connecting two diodes, back to back, which acts as the base terminal.

Statement II : In the study of transistor, the amplification factor β indicates ratio of the collector current to the base current. In the light of the above statements, choose the correct answer from the options given below.

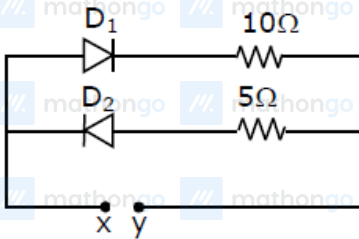
- (1) Statement I is false but Statement II is true.
- (2) Both Statement I and Statement II are true.
- (3) Statement I is true but Statement II is false.
- (4) Both Statement I and Statement II are false.

Q5: 25 Feb (Shift 1) - Single Correct

Questions with Answer Keys

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A 5 V battery is connected across the points X and Y. Assume D_1 and D_2 to be normal silicon diodes. Find the current supplied by the battery if the +ve terminal of the battery is connected to point X.

(1) ~ 0.86 A(2) ~ 0.5 A(3) ~ 0.43 A(4) ~ 1.5 A

Q6: 25 Feb (Shift 2) - Single Correct

Match List I with List II .

List I

- (a) Rectifier
(b) Stabilizer
(c) Transformer
(d) Filter

List II

- (i) Used either for stepping up or stepping down the a.c. Voltage
(ii) Used to convert a.c. voltage into d.c. voltage
(iii) Used to remove any ripple in the rectified output voltage
(iv) Used for constant output voltage even when the input voltage or load current change

Choose the correct answer form the options given below:

(1) (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii)

(2) (a) – (ii), (b) – (iv), (c) – (i), (d) – (iii)

(3) (a) – (ii), (b) – (i), (c) – (iii), (d) – (iv)

(4) (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii)

Q7: 25 Feb (Shift 2) - Single Correct

For extrinsic semiconductors: when doping level is increased;

(1) Fermi-level of p and n -type semiconductors will not be affected.(2) Fermi-level of p -type semiconductors will go downward and Fermi-level of n -type semiconductor will go upward.(3) Fermi-level of both p -type and n -type semiconductors will go upward for $T > T_F K$ and downward for $T < T_F K$, where T_F is Fermi temperature.(4) Fermi-level of p -type semiconductor will go upward and Fermi-level of n -type semiconductors will go downward.

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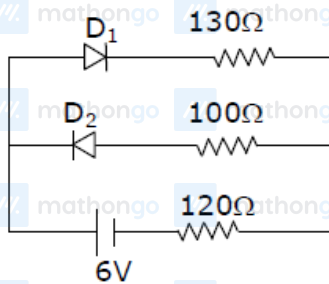
Q8: 26 Feb (Shift 1) - Single Correct

LED is constructed from Ga-As-P semiconducting material. The energy gap of this LED is 1.9eV . Calculate the wavelength of light emitted and its colour. [$h = 6.63 \times 10^{-34}\text{Js}$ and $c = 3 \times 10^8\text{ ms}^{-1}$]

- (1) 654nm and red colour
- (2) 1046nm and blue colour
- (3) 1046nm and red colour
- (4) 654nm and orange colour

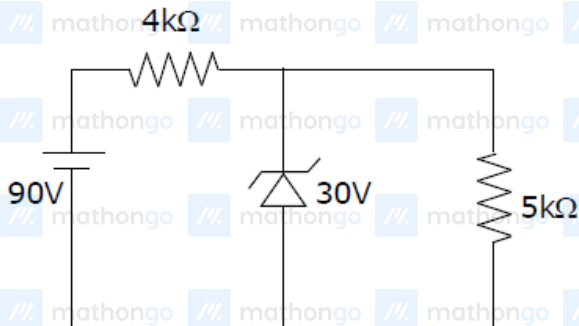
Q9: 26 Feb (Shift 1) - Numerical

The circuit contains two diodes each with a forward resistance of 50Ω and with infinite reverse resistance. If the battery voltage is 6 V , the current through the 120Ω resistance is ____ mA



Q10: 26 Feb (Shift 2) - Numerical

The zener diode has a $V_z = 30\text{ V}$. The current passing through the diode for the following circuit is ____ mA



Answer Key

Q1 (1)	Q2 (25)	Q3 (2)	Q4 (1)
Q5 (3)	Q6 (2)	Q7 (2)	Q8 (1)
Q9 (20)	Q10 (9)		

Q1 (1)

Given :

$$\Delta I_E = 4 \text{ mA}$$

$$\Delta I_C = 3.5 \text{ mA}$$

$$\text{we know that, } \alpha = \frac{\Delta I_C}{\Delta I_E}$$

$$\Rightarrow \alpha = \frac{3.5}{4} = \frac{7}{8}$$

$$\text{Also, } \beta = \frac{\alpha}{1-\alpha}, \text{ so}$$

$$\beta = \frac{\frac{7}{8}}{1-\frac{7}{8}} = \frac{7}{1}$$

$$\beta = 7$$

Q2 (25)

In zener diode there will be no change in current after 5 V

zener diode breakdown

$$\Rightarrow i = \frac{5}{2 \times 10^3}$$

$$\Rightarrow i = 2.5 \times 10^{-3} \text{ A}$$

$$\Rightarrow i = 25 \times 10^{-4} \text{ A}$$

Q3 (2)

The zener breakdown occurs in the heavily doped p – n junction diode. Heavily doped p – n junction diodes have narrow depletion region.

Q4 (1)

S – 1

Statement 1 is false because in case of two discrete back to back connected diodes, there are four doped regions instead of three and there is nothing that resembles a thin base region between an emitter and a collector.

S – 2

Statement-2 is true, as $\beta = \frac{I_C}{I_B}$

Q5 (3)

Since silicon diode is used so 0.7 Volt is drop across it, only D_1 will conduct so current through cell

$$I = \frac{5-0.7}{10} = 0.43 \text{ A}$$

Q6 (2)

(a) Rectifier:- used to convert a.c voltage into d.c. Voltage.

(b) Stabilizer:- used for constant output voltage even when the input voltage or load current change

(c) Transformer:- used either for stepping up or stepping down the a.c. voltage.

(d) Filter:- used to remove any ripple in the rectified output voltage.

Q7 (2)

In n-type semiconductor pentavalent impurity is added. Each pentavalent impurity donates a free electron. So the Fermi-level of n-type semiconductor will go upward.

& In p-type semiconductor trivalent impurity is added. Each trivalent impurity creates a hole in the valence band. So the Fermi-level of p-type semiconductor will go downward.

Q8 (1)

We know that $E = \frac{hc}{\lambda}$

$$= \frac{hc}{E} \Rightarrow \frac{1240(\text{ineV})}{E(\text{ineV})}$$

$$= \frac{1240}{1.9}$$

$$= 652.63\text{nm} \approx 654\text{nm}$$

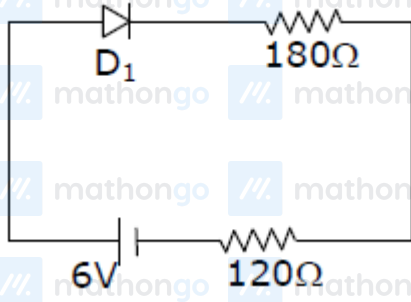
Wavelength of red light is 620nm to 750nm

So, answer is 1.

Q9 (20)

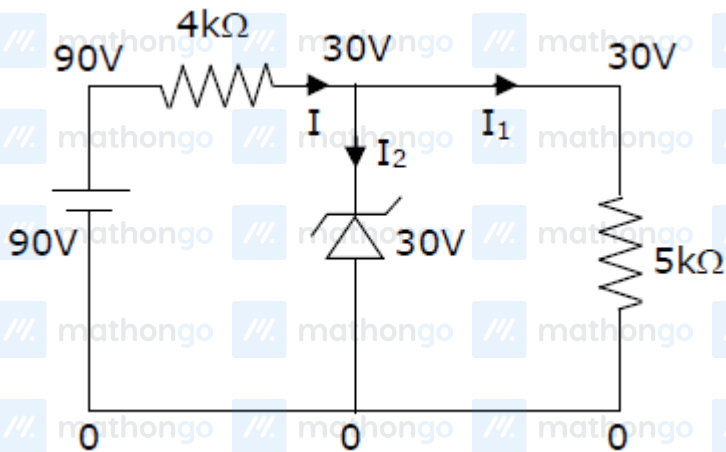
D_2 is reverse bias so current does not flow through D_2

D_1 is forward bias.



$$I = \frac{6}{300} \Rightarrow 0.02 \text{ A}$$
$$= 20 \text{ mA}$$

Q10 (9)



$$I = \frac{90-30}{4} = 15 \text{ mA}$$

$$I_1 = \frac{30}{5 \text{ k}\Omega} = 6 \text{ mA}$$

$$I_2 = 15 \text{ mA} - 6 \text{ mA} = 9 \text{ mA}$$